

**REALTIME MONITORING FRAMEWORK AND ALERTING
SYSTEM FOR UNDERGROUND COALMINE USING
ARDUINO UNO**

A MINIPROJECT REPORT

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**IFET COLLEGE OF ENGINEERING
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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Today safety of miners is a major challenge. Miner's health and life is vulnerable to several critical issues, which includes not only the working environment, but also the after effect of it. Miner's health is in danger mainly because of the toxic gases which are very often released in underground mines. These gases cannot be detected easily by human senses. Our investigates the presence of toxic gases in critical regions and their effects on miners. A real time monitoring system using wireless sensor network, which includes multiple sensors, is developed. This project is using Arduino uno, a microcontroller which acts as a brain of this project. A Bluetooth channel is used to transmit data to create a coal mine safety system. The purpose of this project is to monitor a number of characteristics within the coal mines including fire detection, temperature levels and gas leak detection. This sensor system, installed in the coal mines detect the various parameters in the coal mines. The thinking processors automatically receive all of the sensor rankings, which produces a plethora of ideas for them to consider. The gas monitoring apparatus remains operational to identify potential problems, and a bell serves as a alert when something abnormal happens.

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LIST OF ABBREVIATIONS

UNO	Italian word for ‘ONE
LCD	Liquid Crystal Display
IR	Infrared Radiation
GND	Ground supply
IOT	Internet Of Things
LGPL	Lesser General Public License
LM35	Linear Model 35
IC	Integrated chip
MQ-2	Metal Oxide Semiconductor type gas sensor
SnO2	Silicon dioxide
PC	Personal Computer
MCU	Microcontroller Unit

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Underground mining, especially in coal mines, poses significant challenges to the safety and health of workers. The extraction of coal involves various techniques, and as mines reach greater depths, the risks escalate. Safety concerns are particularly grave due to issues such as inadequate ventilation and the potential for collapses. The deeper the mine, the greater is the risk. These safety issues are of grave concern especially in case of coal industries. Thus, safety of workers should always be of major consideration in any form of mining, whether it is coal or any other minerals. Underground coal mining involves a higher risk than open pit mining due to the problems of ventilation and potential for collapse. However, the utilization of heavy machinery and the methods performed during excavations result into safety risks in all types of mining. Modern mines often implement several safety procedures, education and training for workers, health and safety standards, which lead to substantial changes and improvements and safety level both in opencast and underground mining. Coal has always been the primary resource of energy in India, which has significantly contributed to the rapid industrial development of the country. About 70% of the power generation is dependent on it thus, the importance of coal in energy sector is indispensable. But the production brings with it the other byproducts, which proves to be a potential threat to the environment and the people associated with it. While modern mines implement safety procedures and standards, the need for real-time monitoring systems to detect potential hazards remains paramount. Our project is a small scale initiative aimed at revolutionizing safety practices in

underground coal mining. Acknowledging the inherent risks associated with mining operations, particularly in the coal industry, this project introduces an innovative solution designed to continuously monitor crucial environmental parameters and enhance the safety of miners. The project methodology involves the seamless integration of sensors with Arduino Uno, employing advanced algorithms for real-time data processing. The Bluetooth module enhances the system's accessibility and flexibility, allowing for wireless communication and remote control. Rigorous testing and validation will ensure the reliability and effectiveness of the monitoring system before implementation in real-world underground mining environments. In summary, this project represents a critical step towards advancing safety practices in underground coal mining. By leveraging state-of-the-art technology and a holistic monitoring approach, it aims to significantly mitigate risks, safeguarding the well-being of miners and contributing to the sustainable development of the mining industry. In lieu of that the present work is a sincere attempt in analyzing the graveness and designing a real time monitoring system of detection by using Arduino uno. These small sized sensors are quite inexpensive compared to traditional sensors and also require limited computing and processing resources. These sensor nodes can detect, measure and collect information from the environment and based on some local statistical decision process, they can convey the collected data to the control room.

1.2 OBJECTIVE

The primary objective of the Real-Time Mine Safety System is

- To enhance safety measures in underground coal mines through the implementation of a robust monitoring and alerting system.
- To Develop a system to continuously monitor underground conditions, focusing on temperature and gas levels.

- To implement real-time alerts through audible signals (buzzer) and visual warnings (LCD display) when safety thresholds are exceeded.
- To Enable remote access to real-time mine data via a mobile app, empowering personnel to monitor conditions from a distance.

1.3 ORGANIZATION OF THE PROJECT REPORT

Chapter 1: Gives an introduction and describes the objective of the project.

Chapter 2: Explains the proposed design of the project with block diagram, circuit and the model.

Chapter 3: Give the detailed description of the hardware components used in this project.

Chapter 4: Explains the application of this project.

Chapter 5: Contains the conclusion.

CHAPTER 2

PROPOSED DESIGN

2.1 BLOCK DIAGRAM

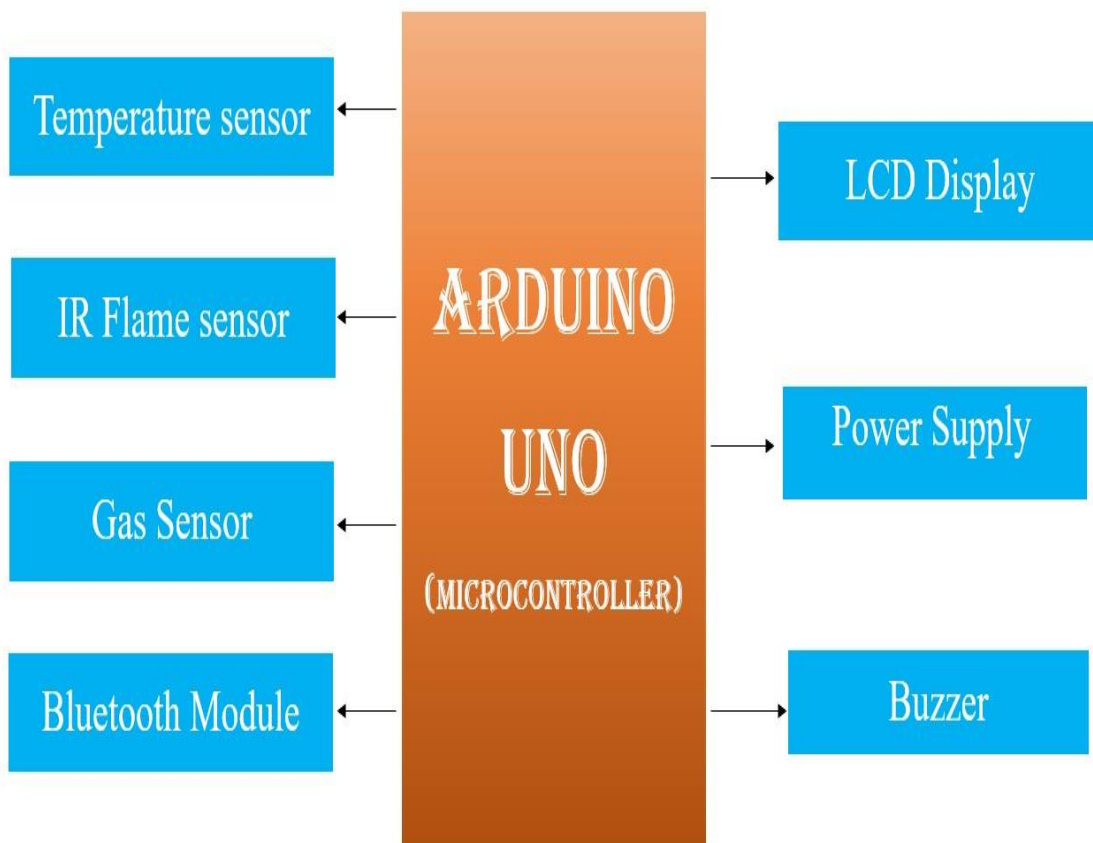


Fig 2.1 Block diagram

The system initializes when the Arduino Uno is powered up. The LCD display shows the temperature, fire and gas values. The temperature sensor and gas sensor continuously monitor their respective environmental parameters. The buzzer is triggered if predefined safety thresholds for temperature, gas levels or fire level are exceeded. Real-time data is sent to a central system via Bluetooth, enabling remote monitoring. Respective Mine personnel receive alerts through a mobile app.

2.2 CIRCUIT OPERATION

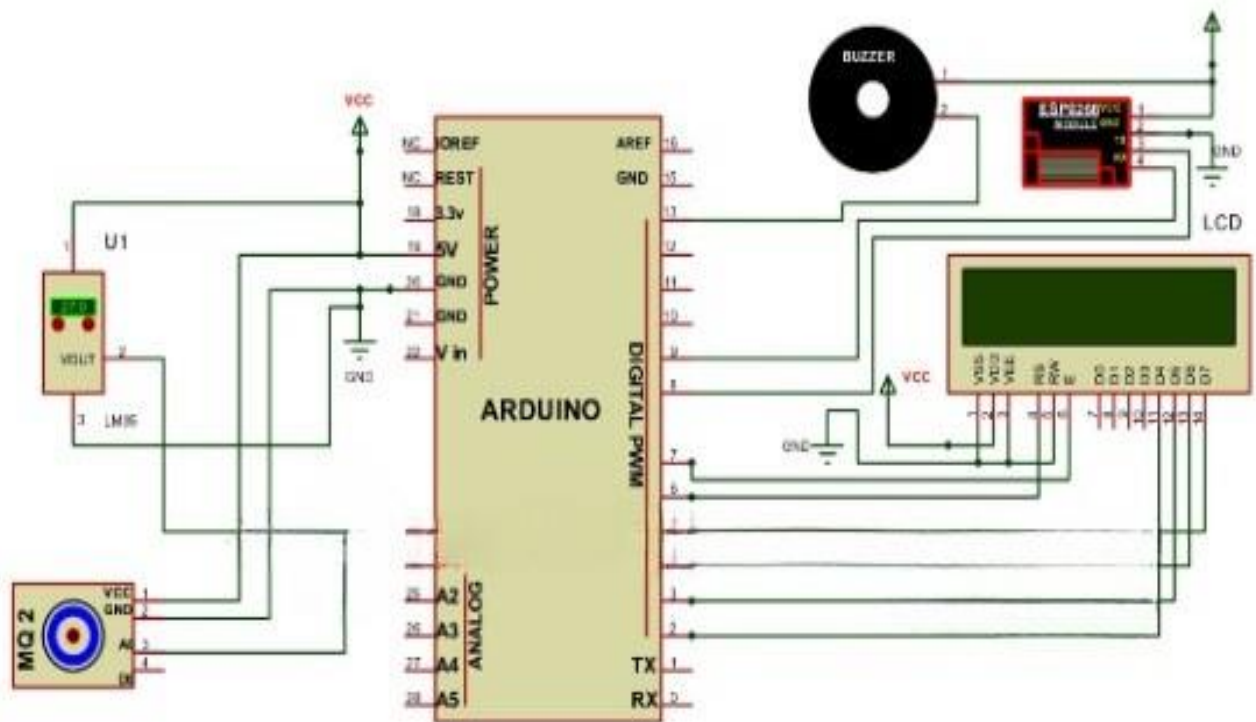


Fig 2.2 Circuit Diagram

The circuit is powered by connecting the VCC and GND of the temperature and gas sensors to the 5V and GND pins on the Arduino Uno. The temperature sensor constantly monitors underground temperature, sending an analog signal to Arduino Uno (connected to, for example, analog pin A0). Similarly, the gas sensor detects hazardous gases and sends an analog signal to Arduino Uno (connected to, for example, analog pin A1). Arduino processes the analog signals from the temperature and gas sensors. If temperature or gas levels exceed predefined safety thresholds, Arduino triggers the alert mechanism. When thresholds are exceeded, the system activates an audible alert through a buzzer. Simultaneously, a visual warning is displayed on the LCD screen. The Bluetooth module is connected to the Arduino Uno, facilitating wireless communication.

2.3 PROPOSED MODEL

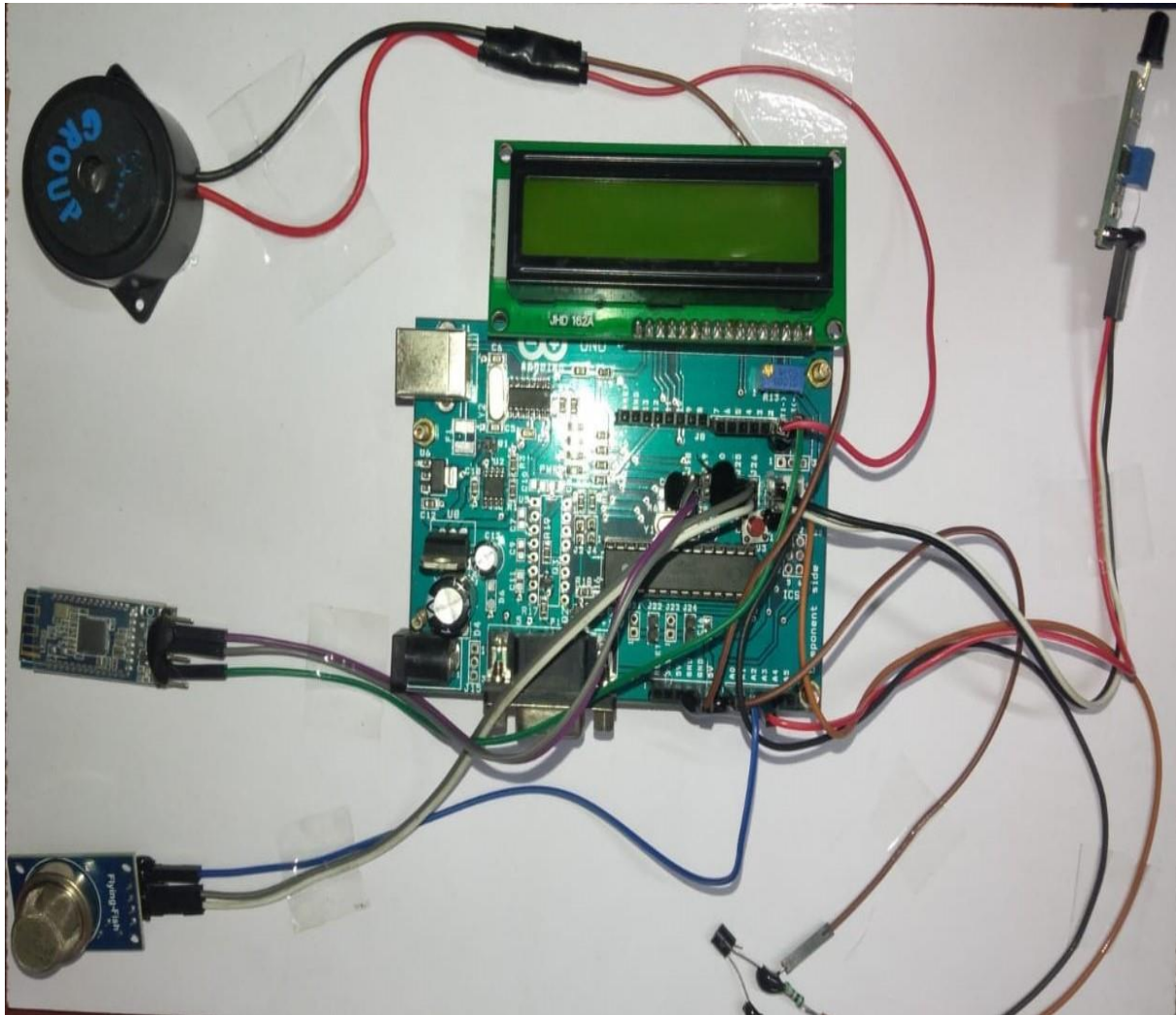


Fig 2.3 Proposed Model

This system we put in place is built on the internet of things (IOT) for coal mine safety. The ambient temperature is detected by a temperature sensor, and if the sensor senses high temperature, the website will be automatically modified, and a buzzer will sound. Additionally, a gas sensor is used to detect the leakage of hazardous gas. The flame sensor detects if any flame is created when the temperature is high. The Bluetooth module sends the data collected wirelessly to the system. If any of the sensors is found to be malfunctioning, the page will be modified automatically.

2.4 SOURCE CODE

```
#include <Wire.h>

#include <Adafruit_MLX90614.h>

#include <SoftwareSerial.h>

#define GAS_SENSOR_PIN A0

#define IR_FLAME_SENSOR_PIN 2

SoftwareSerial bluetooth(10, 11);

void setup() {

    Serial.begin(9600); // Initialize serial communication

    bluetooth.begin(9600); // Initialize Bluetooth communication

    mlx.begin();

    pinMode(GAS_SENSOR_PIN, INPUT);

    pinMode(IR_FLAME_SENSOR_PIN, INPUT);

}

void loop() {

    float temperature = mlx.readObjectTempC();

    int gasLevel = analogRead(GAS_SENSOR_PIN);

    int flameStatus = digitalRead(IR_FLAME_SENSOR_PIN);

    bluetooth.print("Temperature: ");

    bluetooth.println(temperature);

    bluetooth.print("Gas Level: ");

    bluetooth.println(gasLevel);
```

```
bluetooth.print("Flame Status: ");  
  
bluetooth.println(flameStatus == HIGH ? "Fire detected" : "No fire");  
  
Serial.print("Temperature: ");  
  
Serial.println(temperature);  
  
Serial.print("Gas Level: ");  
  
Serial.println(gasLevel);  
  
Serial.print("Flame Status: ");  
  
Serial.println(flameStatus == HIGH ? "Fire detected" : "No fire");  
  
delay(5000); // Delay for 5 seconds before the next reading  
  
}
```


CHAPTER 3

HARDWARE AND COMPONENTS

3.1 ARDUINO UNO

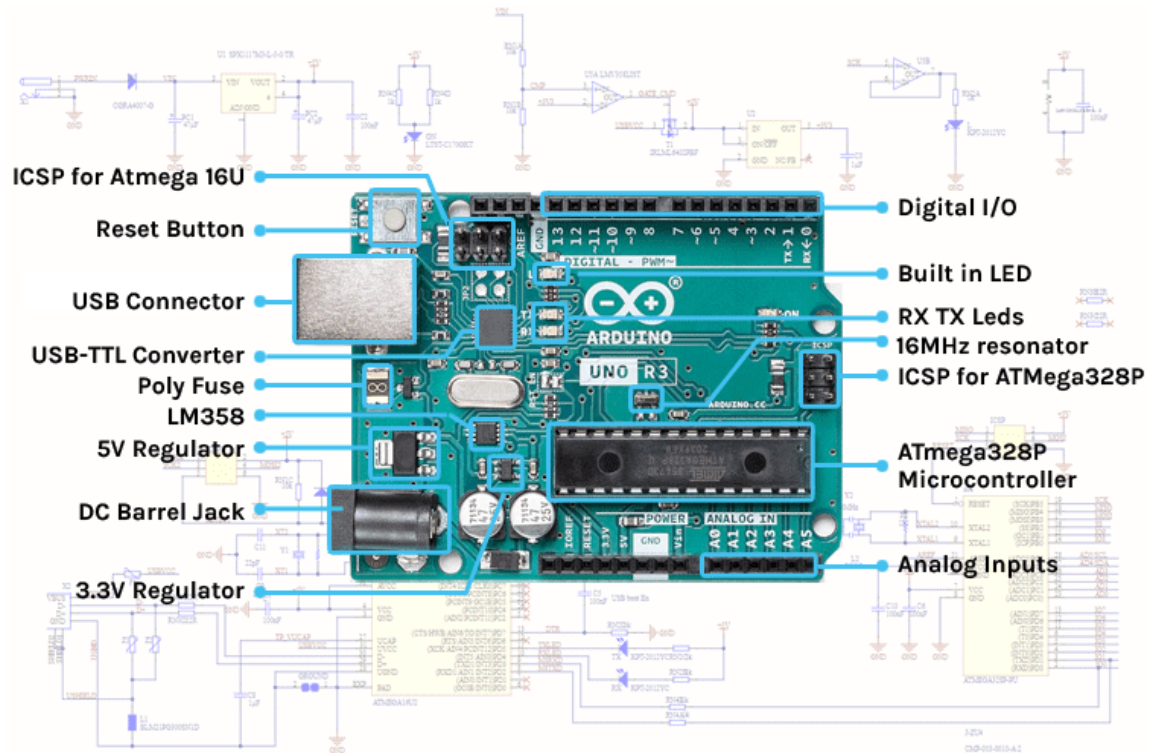


Fig 3.1 Arduino UNO

Arduino is an open-source hardware and software company, project, and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices. Its hardware products are licensed under a CC BY-SA license, while software is licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL), permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially from the official website or through authorized distributors. Arduino is an open-source hardware and software company, project, and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices.

3.2TEMPERATURE SENSOR

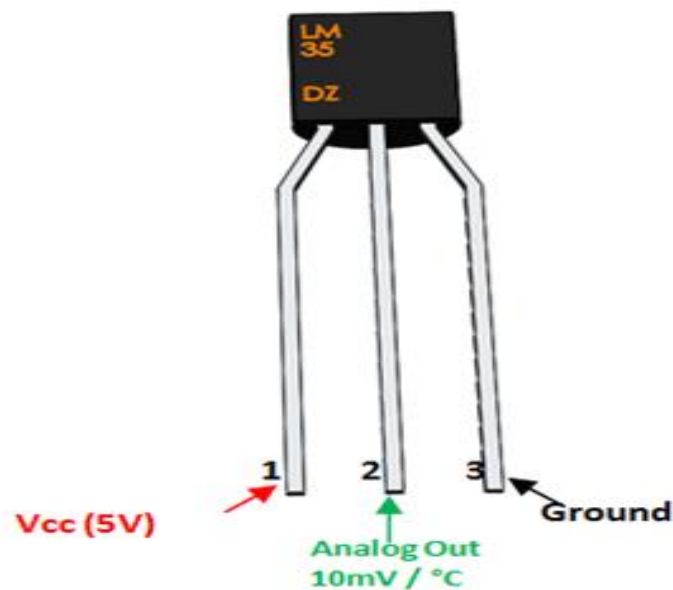


Fig3.2Temperature sensor

The LM35 operates on the principle of the relationship between temperature and voltage. It produces a linear output voltage that is directly proportional to the Celsius temperature it measures. The LM35 is known for its high accuracy, with low self-heating. This makes it suitable for applications where precise temperature measurements are crucial. The sensor provides an output voltage with a sensitivity of 10 Mv/°C. This allows for easy conversion of the analog signal to temperature readings. The LM35 can measure temperatures in a wide range, from -55°C to 150°C, making it versatile for various environments and applications. The LM35 operates on a single power supply in the range of 4 to 30 volts. Its low power consumption makes it suitable for battery-powered and low-energy applications. One of the advantages of the LM35 is its simplicity. It doesn't require external calibration or trimming, providing users with a straightforward temperature reading in degrees Celsius.

3.3 GAS SENSOR(MQ-2)

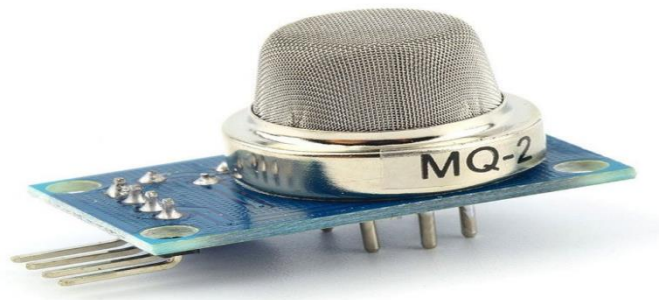


Fig 3.3 Gas Sensor(MQ-2)

MQ-2 is a smoke and combustible gas sensor from Winsen. It can detect flammable gas in a range of 300 – 10000ppm. The gas sensitive material used in the MQ-2 gas sensor is SnO_2 , a low electrically conductive material in clean air. When there is combustible gas in the surrounding air, the electrical conductivity of the sensor will increase with the higher intensity of the combustible gas. Here, we can convert the changing electrical conductivity into output signal by building a simple circuit.

3.4 IR FLAME SENSOR



Fig 3.4 IR Flame Sensor

A sensor which is most sensitive to a normal light is known as a flame sensor. A flame-sensor is one kind of detector which is mainly designed for detecting as well as responding to the occurrence of a fire or flame. The

response of these sensors is faster as well as more accurate. Responses to a detected flame depend on the installation, this project has been included sounding an alarm when a flame is detected. This sensor/detector can be built with an electronic circuit using a receiver like electromagnetic radiation. This sensor uses the infrared flame flash method, which allows the sensor to work through a coating of oil, dust, water vapor, otherwise ice. The main purpose of this sensor is to reduce the risks which are associated with the ignition. These sensors respond frequently than a smoke or heat detector.

3.5 BLUETOOTH MODULE(HC-05)

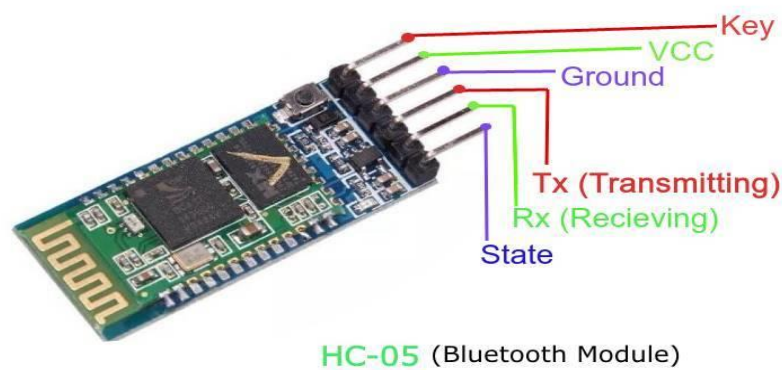


Fig 3.5 Bluetooth Module(HC-05)

The HC-05 is a class 2 Bluetooth module designed for transparent wireless serial communication. It is pre-configured as a slave Bluetooth device. Once it is paired to a master Bluetooth device such as PC, smart phones and tablet, its operation becomes transparent to the user. All data received through the serial input is immediately transmitted over the air. When the module receives wireless data, it is sent out through the serial interface exactly at it is received. Drop-in replacement for wired serial connections, transparent usage. It can be used simply for a serial port replacement to establish connection between MCU, PC to our embedded project and etc.

3.6 LCD DISPLAY



Fig 3.6 LCD Display

A liquid crystal display (LCD) has liquid crystal material sandwiched between two sheets of glass. Without any voltage applied between transparent electrodes, liquid crystal molecules are aligned in parallel with the glass surface. LCDs are commonly used for portable electronic games, in video projection system and in flat-panel televisions.

3.7 PIEZO BUZZER



Fig 3.7 Piezo Buzzer

A piezo buzzer is a type of electronic device that's used to produce a tone, alarm or sound. It's lightweight with a simple construction, and it's typically a low-cost product. Yet at the same time, depending on the piezo ceramic buzzer specifications, it's also reliable and can be constructed in a wide range of sizes that work across varying frequencies to produce different sound .

CHAPTER 4

RESULTS AND DISCUSSION

4.1 OUTPUT

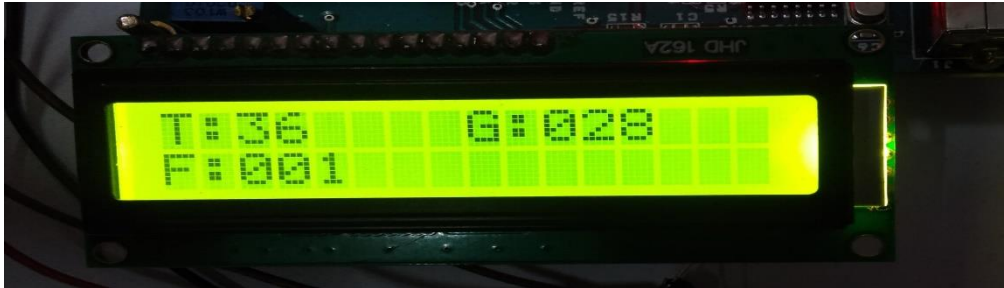


Fig 4.1 Safe Condition



Fig 4.2 Gas Alert



Fig 4.3 Fire Alert



Fig 4.4 Temperature Alert

4.2 IOT MONITORING

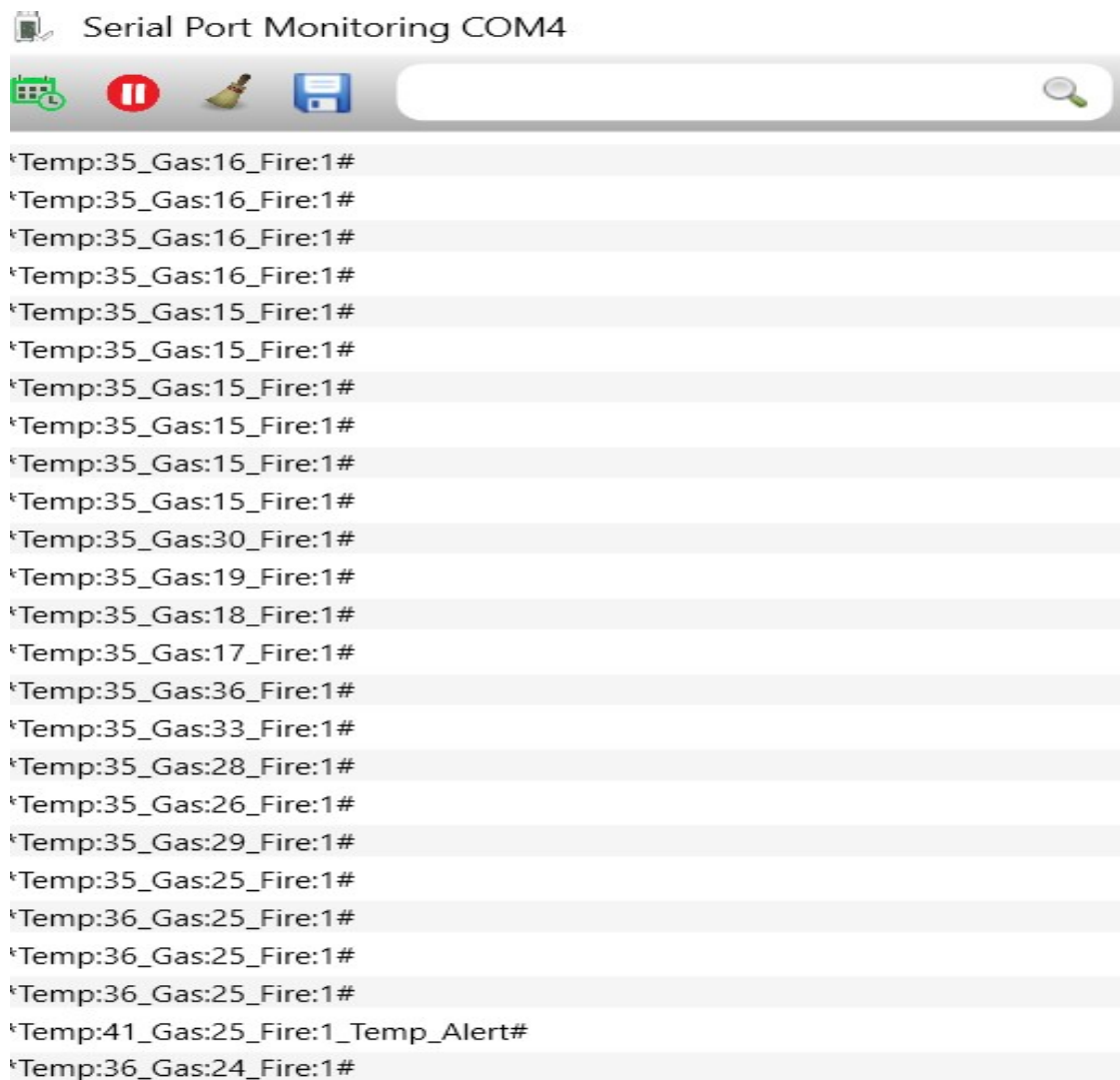


Fig 4.5 Realtime Monitoring

- **Temperature Sensor:**

The MLX90614 infrared temperature sensor successfully recorded temperature changes within the cardboard-simulated underground environment.

- **Gas Sensor:**

The gas sensor, placed within the cardboard setup, effectively monitored simulated gas levels. The analog readings, translated into parts per million (PPM).

- **IR Flame Sensor:**

The IR flame sensor, positioned strategically within the cardboard mine, reliably indicated the presence or absence of flames. The digital output accurately reflected the status of flame detection.

- **Bluetooth Module:**

The Bluetooth module, integrated into the Hardware setup, facilitated wireless communication. Real-time data, including temperature, simulated gas levels, and flame detection status, was successfully transmitted to external devices.

CHAPTER 5

APPLICATIONS

- The system can be deployed in underground coal mines to detect and monitor hazardous conditions such as gas emissions, temperature anomalies, and potential fire incidents in real-time.
- The Bluetooth module enables wireless communication, allowing data retrieval and system control from a remote location.
- This application is particularly useful for accessing real-time information without miners having to physically be in the mine.
- Implementation of this system contributes to overall safety enhancement in mining operations.
- By continuously monitoring critical parameters, the system provides valuable data for preemptive safety measures, reducing the risk of accidents.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 CONCLUSION

Wireless sensor networks have been used to examine data from a study on the real-time monitoring of poisonous gasses and other factors found in underground mines. To give a more precise and lucid view of the underground mine, a real-time monitoring system is being built. This system will allow all miners inside the mine to save their lives before any casualties occur by displaying the parameters on the LCD at the underground section where the sensor unit is located as well as on the monitoring unit. When sensor values exceed the threshold, an alarm is set off. Additionally, this system keeps all of the data on file for later review. An extra and more defined goal is provided for evaluating mine point perspective with the installation of a real-time observation device, increasing accuracy. This method will in this instance show the parameters on the monitoring screen. This plan will benefit all present mine workers as well as other workers, as they will be able to use it to save themselves from dying in a work-related accident.

6.2 FUTURE SCOPE

Explore the integration of additional sensors for a more comprehensive environmental analysis, such as air quality, humidity, or seismic activity. Develop an autonomous response system that can take immediate safety actions without manual intervention in critical situations. Improve the user interface of the mobile app for more intuitive navigation and detailed insights into mine conditions. Design the system to be scalable and adaptable for use in different types of underground mines, considering variations in geological and operational conditions.

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